



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत

Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in

B.Tech. Second Sem End Semester Examination, 14th MAY, 2025.

Roll No...12...12003

Subject: Engineering Mathematics-II

Subject Code: MAL201

Branch: IT

Max.Marks.-50

Time- 3 Hours

Note: All questions are compulsory. (Rough sketch compulsory in Q-1, 2 (a) and 2(b)).

Q-1	Using Method of Change of Variables, Find the volume of the portion of the sphere $x^2 + y^2 + z^2 = b^2$ lying inside the cylinder $x^2 + y^2 = bx$.	5 Marks
Q-2	(a) By changing the order of integration and hence evaluate $\int_0^a \int_{y^2/a}^y \frac{y}{(a-x)\sqrt{ax-y^2}} dx dy$. (b) Transform the double integral to polar coordinate and hence evaluate it, $\int_{y=0}^{y=a} \int_{x=\sqrt{a^2-y^2}}^{x=\sqrt{a^2-y^2}} (a^2 - x^2 - y^2) dx dy$.	3 Marks 2 Marks
Q-3	Using Picard's iteration Method find the First two Approximation to the solution of the initial value problem $y' = x^2 + y^2$ such that $y(0) = 1$, Also Compare with exact solution.	5 Marks
Q-4	(a) Determine the value of 'm' such that the curve $x^m + y^m = mC$ are orthogonal trajectories of the family of curves $y = \frac{x}{1-xC_2}$, where 'C' is the constant of integration. (b) Solve the Differential Equation Completely: $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$.	3 Marks 2 Marks
Q-5	Find the General Solution of the Legendre Linear Ordinary Differential Equation, $((x+1)^2 D^2 - 3(x+1)D + 1)y = \left\{ \frac{\sin(\log(x+1))+1}{(x+1)} \right\} \log(x+1)$.	5 Marks
Q-6	(a) Solve the Following Linear Differential Equations for $x(t)$ & $y(t)$, $(D-2)x + (2D-1)y = e^{2t}$, $x + (D-2)y = 0$ and also find order of determinant. (b) Solve the ODE: $(D^2 + 4)y = \tan 2x$ By Using Method of Variation of Parameters.	3 Marks 2 Marks
Q-7	Find the Complementary Function and Particular Integral of the following PDE, $(D^2 - D'^2)z = e^{x-y} \sin(x+2y)$	5 Marks
Q-8	(a) Solve the following Partial Differential Equation: $px + qy = pq$ By Charpit's Method. (b) Find the complete solution of PDE: $xp^2 + yq^2 = z$, where $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$.	3 Marks 2 Marks
Q-9	Using Laplace Transform Method Solve the Differential Equation: $y'' + y = e^{-2t} \sin t$ that satisfies the initial conditions $y(0) = 0$ and $y'(0) = 0$.	5 Marks
Q-10	(a) Use the Definition of Laplace transform, Evaluate the integral $\int_0^\infty e^{-2t} \cdot \left(\int_0^t e^{ct} \left(\frac{\cos at - \cos bt}{t} \right) dt \right) dt$. (b) Using Convolution Theorem find inverse Laplace Transform $L^{-1} \left[\frac{s^2}{(s^2+1)^2} \right]$	3 Marks 2 Marks



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End Sem

Subject: OOPS Using C++(UCS004)

Roll no .12.4.1.2003

Branch: IT/CSE

M.M.-50 marks

Time- 3 Hrs.

Note: All questions are compulsory.

Oops concept can be used as per requirements in Q6 and Q7: Templates, File Management, Exception Handling, Classes, Pointers, Arrays, Inheritance, Polymorphism, Encapsulation, Abstraction.

Question No.	Questions	Marks
Q-1:	Compare static binding and dynamic binding in C++. When would each be appropriate?	4
Q-2:	How can constructors and destructors be analyzed to ensure proper resource management in a class?	4
Q-3:	Critique the design choice of making all data members public in a class.	4
Q-4:	Compare deep copy and shallow copy using a C++ example.	4
Q-5:	Given a program that heavily uses global variables and static functions, assess its object-oriented design quality.	4
Q-6:	IIIT Sonepat need to develop an inventory system for a warehouse to track products, having different product categories (e.g., Computer, Laptop, Furniture, etc). Use templates for product IDs. Save inventory data to a file, Inheritance for product types, Polymorphism for stock updates and handle errors like negative quantities. Elaborate the design in accordance to class name, data type, and member function along with code.	15
Q-7:	IIIT Sonepat needs to develop a system to manage student results including different result types (e.g., Mid Term1, Mid-Term2, End semester). Use oops concept by applying templates for (student Ids, course code and name), file handling to save the results to a file, Pointers for student objects, Polymorphism for grade calculation and handle errors like invalid marks including Abstraction. Elaborate the design in accordance to class name, data type, and member function along with code.	15



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2nd Semester End Term

Subject: Web Designing (UCS003)

Roll no: 12412003

Branch: CSE/IT/DSA

M.M.-50 marks

Time- 180 minutes

All Questions Are Compulsory!

Q-1	You are tasked with creating a simple, informative web page for a technology seminar on "The Evolution of the Internet and Web Technologies." Your page must demonstrate your understanding of basic web concepts and practical coding skills. a) Briefly explain the difference between client-side and server-side scripting, and provide two examples of technologies used for each. (2M) b) Describe the role of the World Wide Web Consortium (W3C) and its importance in web development. (2M) c) Identify and explain two key features of HTML5 and how they improve web development compared to earlier versions. (2M)	(2x3=6M)
Q-2	You are designing a simple HTML5 page for a restaurant website. Your goal is to demonstrate understanding and application of foundational HTML5 concepts. a) Explain the purpose of the <code>alt</code> attribute in image elements. (1M) b) Describe the difference between ordered, unordered, and <code>(description)</code> lists in HTML5. Provide one appropriate use case for each. (2M) c) Write the HTML5 code for a restaurant's "About Us" page that includes: (2M) - A main heading (<code><h1></code>) and a subheading. - An image of the restaurant (use a placeholder like <code>restaurant.jpg</code>) with alt text. - A paragraph with a special character (e.g., copyright). - A horizontal rule. - An unordered list of 3 dishes. - A table listing 3 dishes and their prices. d) What is the purpose of the <code>meta</code> element in HTML5? Provide two examples of how it is used in the <code><head></code> section. (2M) e) Explain how the W3C HTML5 Validation Service helps improve website quality. Mention one common validation error and how to fix it. (1M) f) Create a simple form that: (2M) - Collects customer's name and email. - Includes a submit button. - Uses <code>placeholder</code> and <code>required</code> attributes.	(10M)
Q-3	You are designing a "User Profile Setup" page for a modern web application. Your task is to implement new HTML5 form input types, enhance user experience with datalist and autocomplete, and structure the page using semantic elements. a) Describe the benefits of using new HTML5 form input types (e.g., email, date, range, color) over traditional text inputs. Explain how they improve: (2M) - User experience - Data validation	(9M)



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2nd Semester End Term

Subject: Web Designing (UCS003)

Roll no: _____

Branch: CSE/IT/DSA

M.M.-50 marks

Time- 180 minutes

(Mobile responsiveness)

b) Explain the purpose and usage of the `datalist` element and the `autocomplete` attribute. Provide an example where both are used together in a form field. (2M)

c) Write the HTML5 code for a user profile form that includes: (3M)

- A text input for full name (with `autocomplete` on)
- An email field
- A date picker for birth date
- A color picker for selecting a favorite color
- A range input for selecting experience level (1-10)
- A `datalist` input for selecting a preferred programming language (e.g., JavaScript, Python, C++)
- A submit button

d) Using semantic HTML5 structure, wrap the form inside a well-structured layout that includes: (2M)

- A `<header>` with a title
- A `<section>` for the form
- An `<aside>` with a `details` element that includes a summary like "What is this form?" and a description
- A `<footer>` with author name and year

Q-4 a) Define the box model in CSS. Identify its four components and explain how it affects the layout of elements on a page. (3M)

(9M)

b) What happens when there are conflicting styles between inline, embedded, and external styles? Describe the CSS (cascade) and specificity rules, and give an example where one style overrides another. (2M)

c) Write the HTML and CSS code (use an embedded style sheet) for a portfolio section that includes: (2M)

- A container `<div>` with a background color and fixed width and height
- An absolutely positioned `<div>` for a profile image with a `z-index` above the rest
- A paragraph styled with padding, border, and margin
- A `<span>` inside the paragraph that is relatively positioned and styled differently (e.g., bold and red)

d) Briefly describe the purpose of media queries in CSS. Write a simple example that changes the page background color when the screen width is less than 600px. (2M)

Q-5 a) Explain the purpose and difference between the following CSS3 properties: (2M)

(6M)

- `box-shadow`
- `border-radius`
- `linear-gradient`



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M.M.-50 marks

Time- 180 minutes

Provide an example scenario where each could be effectively applied on a web page.

b) Write the CSS (embedded or internal style) for a landing page header that includes: (2M)

- A background with a radial gradient
- A heading with a text shadow and rounded corners
- A button that transitions smoothly (e.g., changes color and grows slightly) when hovered
- Use of a Google Font or custom web font for the heading

c) What are CSS transitions and transformations? Briefly explain how the `transform: scale()` and transition properties can work together to create interactive effects. (2M)

Q-6 You are developing a simple web-based game where a user rolls two virtual dice. The result should be displayed, and the user wins if the total equals 7 or 11. The game also tracks the number of times the user has won or lost.

(10M)

a) Briefly explain each of the following JavaScript concepts with a code snippet for each: (3M)

- Getting user input using `prompt()`
- Using `if...else` for decision-making
- Creating a simple function

b) Describe the difference between the following control structures: (2M)

- `while` vs. `do...while`
- `if...else` vs. `switch`

Provide one appropriate use case for each pair.

c) What is the difference between local scope and global scope in JavaScript? Explain with an example showing a variable declared (`var` keyword) inside a function and one outside. (2M)

d) Write a complete JavaScript program that: (3M)

- Defines a function `rollDie()` that returns a random integer from 1 to 6
- Rolls two dice, calculates the sum, and checks if the user wins (sum is 7 or 11)
- Tracks how many games the user has played and how many times they've won or lost



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Roll No.: 12412003

COMMUNICATION SKILLS (HUL 204)
END TERM EXAMINATION MAY/JUNE 2025

Branch: CS/IT/DSA
Semester II Batch 2024-28

M.M.-50marks

Attempt All Questions. All questions carry equal marks.

1. Write an essay on the differences between written and visual communication. Explain visual communication with the help of suitable examples.

(1 x 10= 10)

2. Read the text in the box carefully. Which type of writing is it? Comment on purpose, nature, and effect of this writing on its reader. Also, discuss language structure of the text below.

(1 x 10= 10)

"Silhouetted against a golden moon, which grew larger each moment, was a big, oddly lop-sided creature, which was coming in Harry's direction. He stood very still and watched it sink lower. For a moment, he hesitated with his hand on the latch of the window, deciding whether he should slam it shut, but the strange creature soared over a streetlight of Privet Drive, and then Harry, who had realised what it was, jumped aside."

3. Write a detailed note on non-verbal communication. Discuss its types and explain them with the help of suitable examples.

(1 x 10= 10)

4. Discuss the significance of a thesis statement and topic sentence. Provide suitable examples and elaborate on its features.

(1 x 10= 10)

5. What do you understand by 'barriers to communication'. Explain, with suitable examples, barriers to communication in rapid speech with specific reference to varieties of English.

(1 x 10 = 10)



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End Sem. Exam Subject Name: Data Structure Subject code: UCS 002 Roll no. 12.A12003
Branch: DSA/IT 2th Semester M.M.-50 marks Time- 3 Hours

Note: (1) Attempt all the questions.

Q-1	Describe the properties of trees including height, level, and depth. Explain the implementation of tree traversals (Inorder, Preorder, Postorder) using recursion. (How many unlabeled and labelled Binary Search Trees (BSTs) are possible with 8 nodes.) illustrate the concept of inserting the keys 21, 26, 30, 9, 4, 14, 28, 18, 15, 10, 2, 3, and 7 into an initially empty BST tree.	(10 marks)
Q-2	Construct an AVL Tree by inserting the keys 30, 20, 10, 15, 25, 23, 39, 35, and 43 into an initially empty tree. Then, provide the Preorder, Inorder, and Postorder sequences for the resulting AVL Tree. Finally, explain the process of deleting root nodes in this AVL Tree.	(10 marks)
Q-3	You are given the unsorted array $A = [45, 23, 67, 12, 89, 34, 78]$. Apply Merge Sort on this array and demonstrate each step involved in the division and merging process with subarray representation. Then, perform Quick Sort on the same array using the last element as the pivot and show each partitioning stage. Compare the total number of comparisons made by each algorithm. Conclude which algorithm performs better for this dataset and explain why.	(10 marks)
Q-4	Consider a stack implemented using an array with a maximum size of 10 elements. Initially, the stack is empty. Perform the following sequence of operations on the stack: Push 5, Push 8, Push 3, Pop an element, push 10, Push 7, Pop an element, push 2, Pop an element, push 4, Push 9. At the end of these operations, what is the top element of the stack? Moreover, (how a stack can be implemented using two queues?)	(10 marks)
Q-5	Write a function to delete a node from a singly linked list given a value key. Explain how your code handles the following edge cases: deleting the head, deleting the last node, deleting a middle node, and deleting a non-existent element. Use the list $7 \rightarrow 14 \rightarrow 21 \rightarrow 28 \rightarrow 35 \rightarrow 100$ and demonstrate the deletion of 7, 35, 21, and 100 respectively. Illustrate the list at each step and discuss potential null pointer exceptions and how to avoid them.	(10 marks)

End



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End Sem Exam

Subject: Digital Electronics (ECL-202)

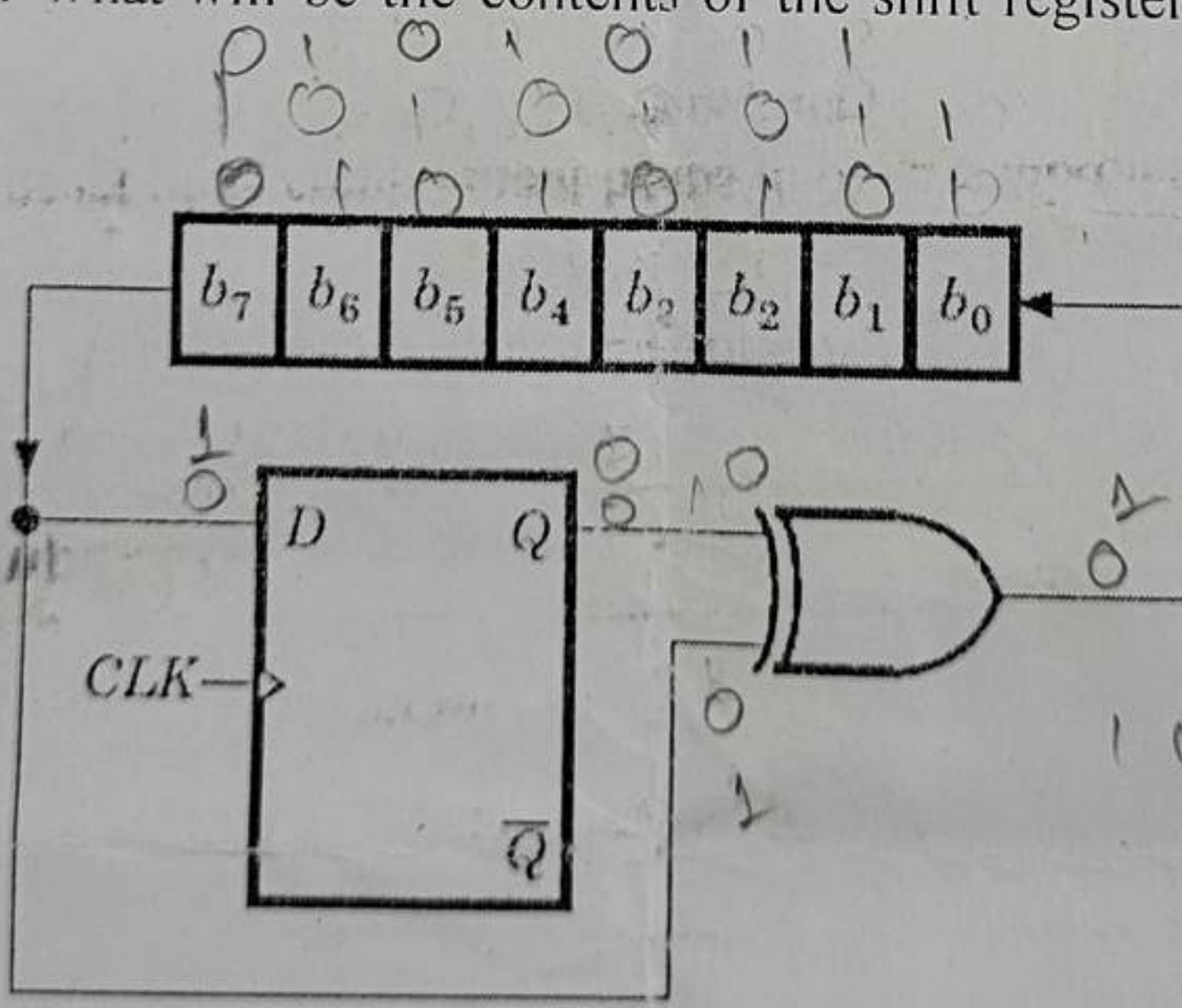
Roll no. 12412003

Branch: CSE/IT

M.M.-50 marks

Time- 3 Hrs.

Note: All questions are compulsory.

Q-1: (a)	What are the differences between standard and canonical form of Boolean algebra? Convert the following expression into both Canonical SOP and Canonical POS forms: $F = A + B'C$	(5 marks)
(b)	Explain how the concept of 1's complement helps in binary subtraction with an example. Subtract 1001 from 0101 using 1's and 2's complement method, and interpret the result.	(5 marks)
Q-2: (a)	Using a 3-to-8 line decoder and an 8-to-1 multiplexer, design a custom combinational circuit that outputs 1 only when the binary input is a prime number. Draw the circuit and describe the logic of interconnection.	(5 marks)
(b)	Design a combinational circuit that converts a 3-bit Gray code to its binary equivalent. Derive the truth table, logic expressions, and circuit diagram.	(5 marks)
Q-3: (a)	Design a synchronous counter using JK flip-flops that counts the number from 0 to 5.	(5 marks)
(b)	The 8-bit left shift register and D-flip-flop shown in below figure is synchronized with same clock. The D flip-flop is initially cleared and the 8-bit register is initialized to value of 01010101. What will be the contents of the shift register after applying five clock pulses. 	(5 marks)
Q-4: (a)	Describe the operation of a Tracking ADC with a block diagram. Compare its performance with a Counter-type ADC, and analyze its response to slowly varying inputs using timing diagrams for increasing and decreasing signals.	(5 marks)
(b)	A 4-bit DAC operates with a reference voltage of 10V. Determine the step size and resolution of this DAC, and calculate the analog output voltage corresponding to the	(5 marks)

	digital input 1001.	
Q-5: (a)	Compare and contrast Static RAM (SRAM) and Dynamic RAM (DRAM) in terms of their internal structure, operation, and performance characteristics.	(5 marks)
(b)	Provide differences between (Any two): i) Volatile and Non-volatile memory ii) Synchronous and asynchronous sequential circuits iii) Combinational and Sequential Circuits	(5 marks)

*****Best Wishes*****